

Introduction

By 2050, the amount of people with diabetes is expected to be at least **1.3 billion** worldwide. **Diabetic neuropathy** is a complication of poorly managed diabetes, where **high blood sugars** over an extended period can damage nerves in the feet. If left unattended, imperceptible cuts and sores can **develop into ulcers** that are slow to heal. These ulcers if infected can lead to last-ditch solutions such as amputation. **Diabetes accounts for 60 - 70% of all lower extremity amputations.**



Fig. 1: A Commercially available in-shoe based foot plantar pressure sensor by Pedar®

It is not very accessible to those in **low-income areas**, which accounts for a higher number of diabetics. These high prices can be attributed to the customization for each patient's foot, and **high-priced materials** for the insoles.

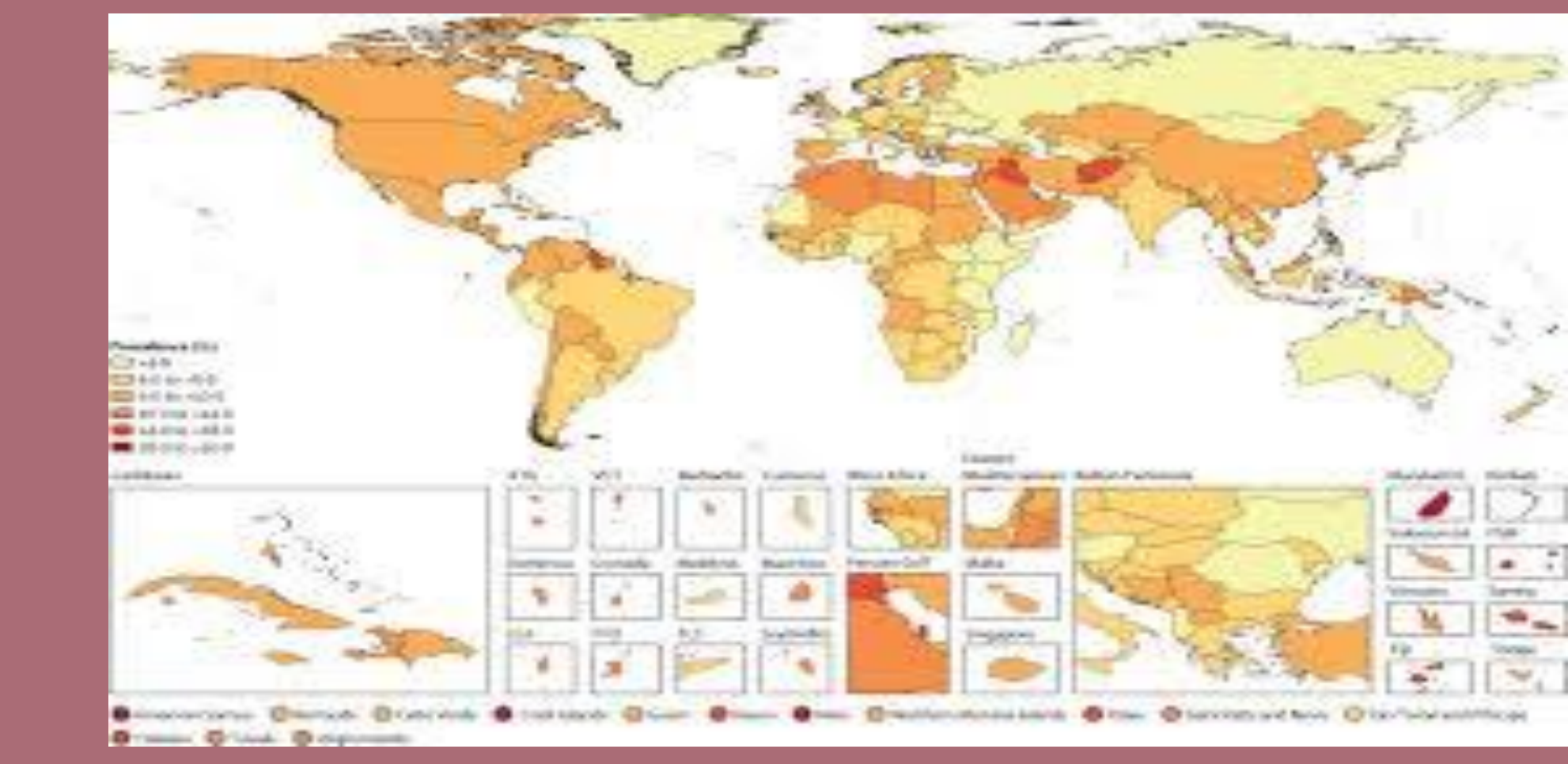


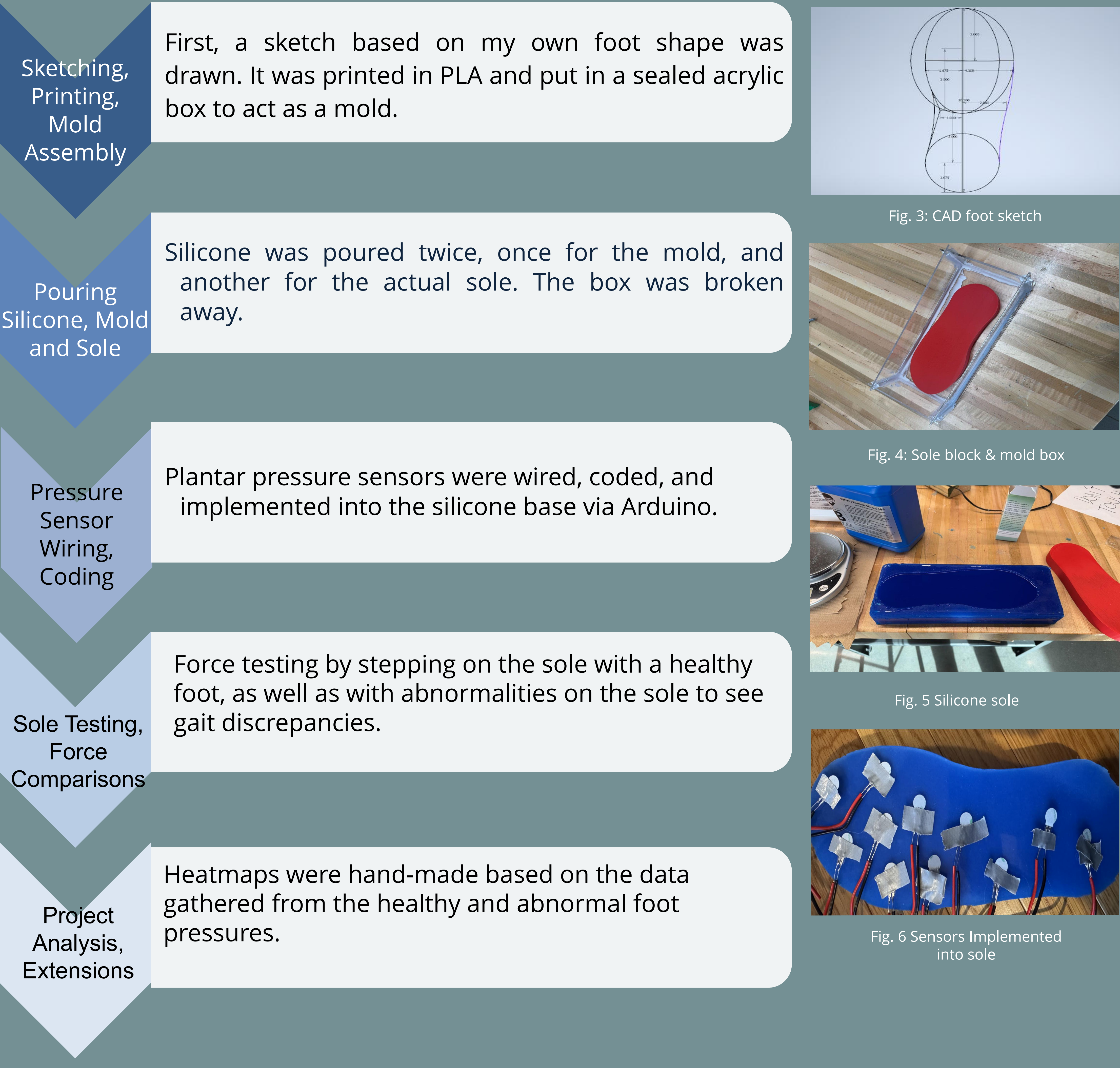
Fig. 2: Distribution map of diabetes diagnoses (The Lancet)

Purpose

Create a **continuous pressure sensor** for individuals with diabetic neuropathy to replace the loss of sensation in their foot, ultimately **preventing amputation**. Furthermore, I aim to make this insole **cheap and accessible**, so that even in lower income areas, the option of plantar pressure mapping is available. To fulfil this, the target price for a functional prototype is under \$200, as commercial solutions can cost over \$10,000.

Detection of Diabetic Foot Ulceration via Plantar Pressure Sensors in Soles

Methodology



Test #	Sensor #										
	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
1	2.54 kg	-	-	-	-	-	2.32 kg	-	-	-	3.40 kg
2	2.89 kg	-	2.19 kg	2.40 kg	-	-	2.27 kg	2.37 kg	-	-	3.25 kg
3	3.03 kg	2.34 kg	2.68 kg	2.22 kg	3.36 kg	2.94 kg	2.41 kg	1.97 kg	3.06 kg	2.74 kg	2.88 kg
4	3.03 kg	2.34 kg	2.68 kg	2.22 kg	3.36 kg	2.94 kg	2.41 kg	1.97 kg	3.06 kg	2.74 kg	2.88 kg

Fig. 7: Table of Test Data

Results

Due to the **unrefined shape** of the sole bed, the prototype's data is unclear. As a constant, the foot with no discrepancy added displayed **varying pressures** across the sole. When the abnormal foot was modeled, very **little change** was noted.

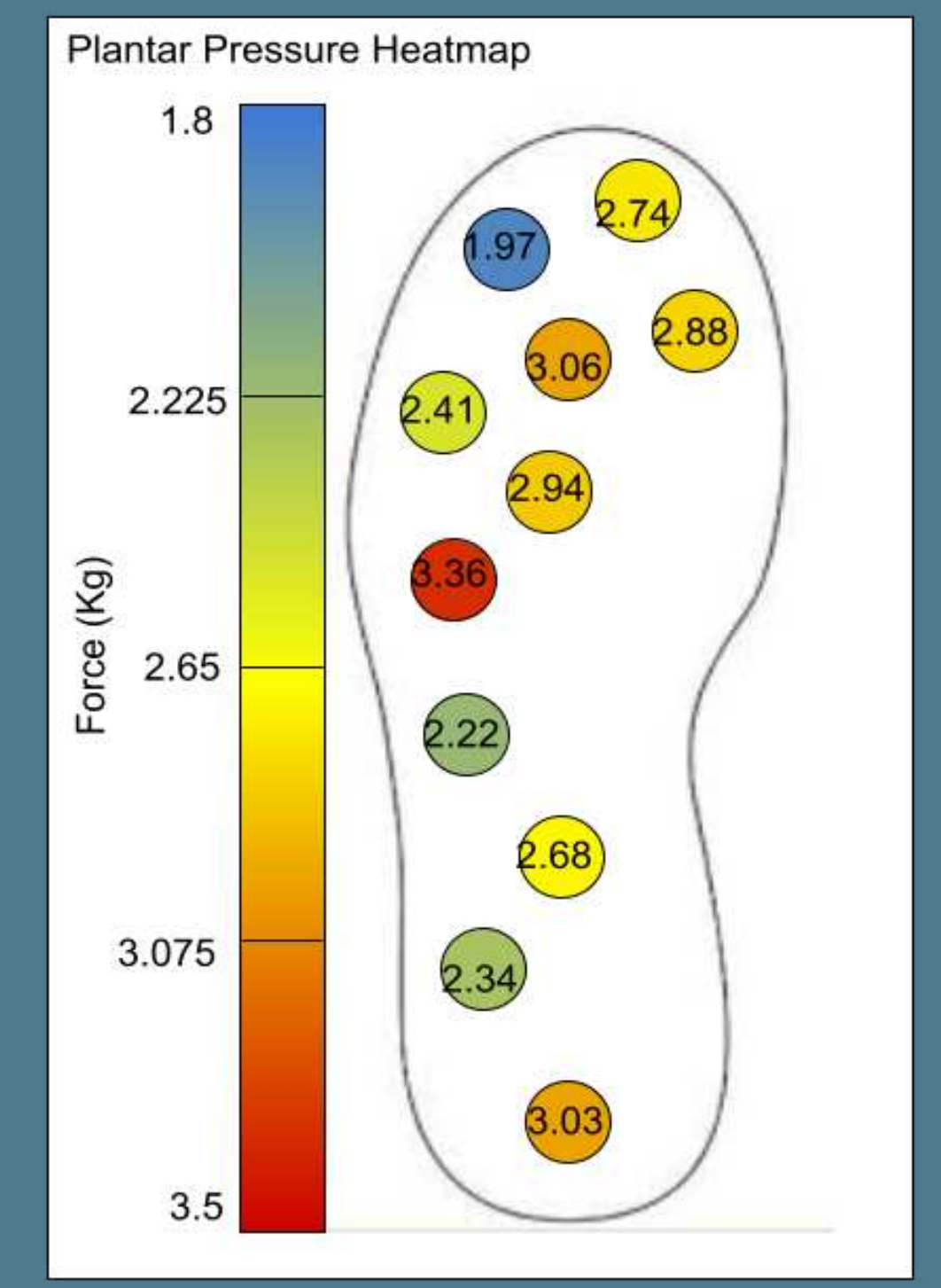


Fig. 8: Pressure Distribution Heat-map of a Healthy Foot.

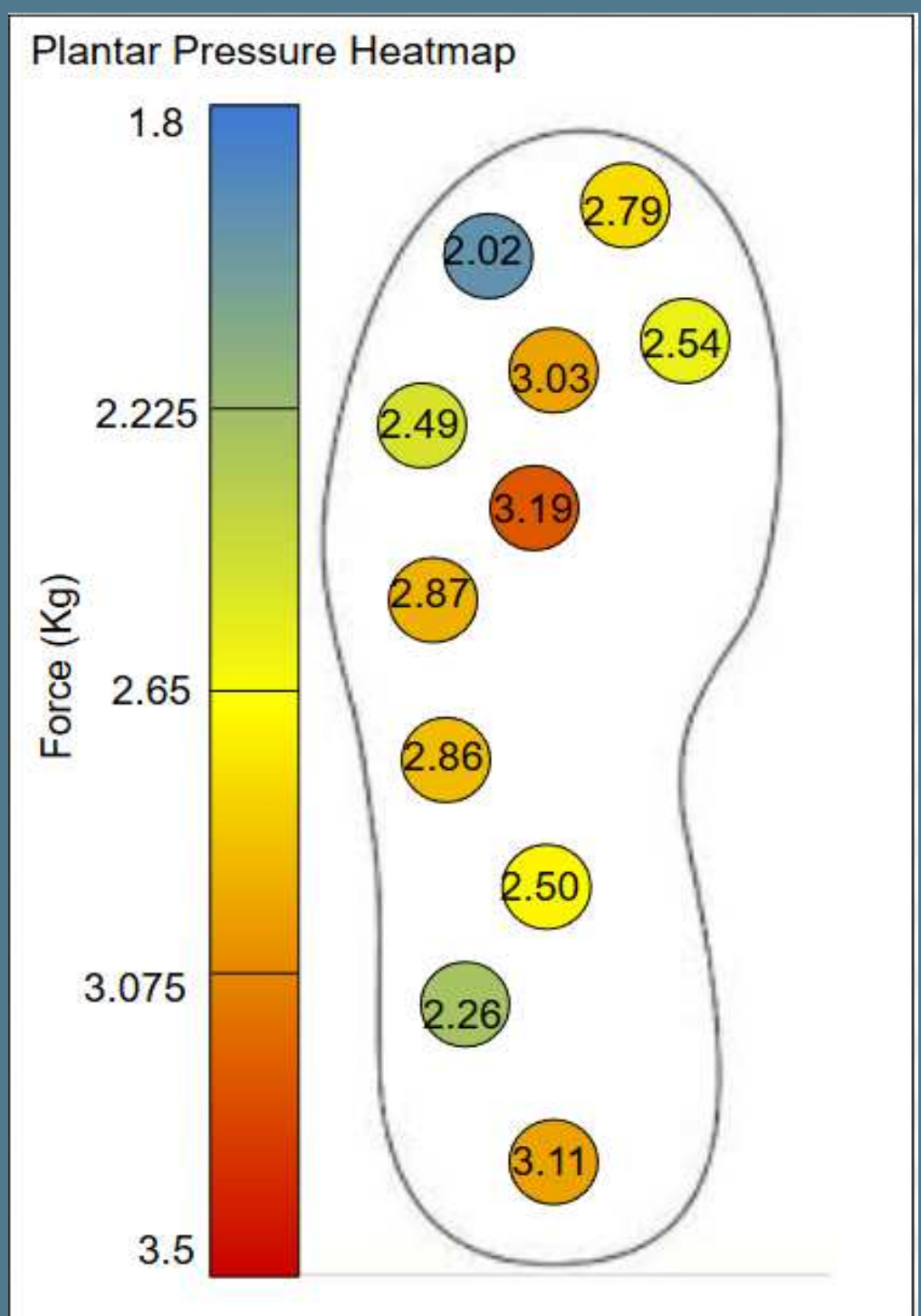


Fig. 9: Pressure Distribution Heat-map of an Abnormal Foot.

Conclusion

In theory this **concept is feasible**. Multiple studies have shown success in ulcer detection through plantar pressure sensors. Even though this prototype was flawed, there is **readable data and the prototype was inexpensive**. A cheaper alternative to ulcer detecting footwear is **possible** via this methodology with **certain alterations**. In future attempts of this study, to prevent the same roadblock, a **different material** should be used for the negative of the sole, such as a poured plaster, and adequate mold release should be applied. Alternatively, the fabrication of a **3d printed, thermoplastic polyurethane** sole would ensure a smooth surface for pressure readings. Either of these applied changes would **improve the clarity** of the data gathered.

Citations

Global, regional, and national burden of diabetes from 1990 to 2021, with projections of prevalence to 2050: A systematic analysis for the global burden of disease study 2021 - the Lancet. (n.d.). [https://www.thelancet.com/article/S0140-6736\(23\)01301-6/fulltext](https://www.thelancet.com/article/S0140-6736(23)01301-6/fulltext)

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